

# Dietary Protein Digestion & Absorption

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# Nitrogen Intake & Digestive Architecture

**70–100g**

Daily Protein Consumption for an  
Adult Man

**3 Organs**

Proteolytic Enzymes produce by  
Stomach, Pancreas & Small Intestine

**Newborns**

Intact Macromolecule Uptake  
Absorb antibodies directly from  
breast milk

**Proteins are primary constituents of the body & are too large for intestinal absorption and must be hydrolyzed into dipeptides, tripeptides, and free amino acids.**

- Dietary proteins are the primary source of **nitrogen**.
- A regular & adequate supply of protein is essential for **cell integrity & function**.

# Proteins provide many essential functions in the body

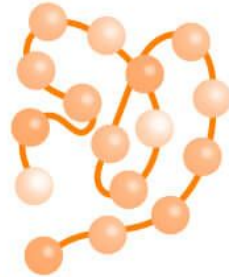


## Proteins help create digestive enzymes

Digestive enzymes help facilitate chemical reactions

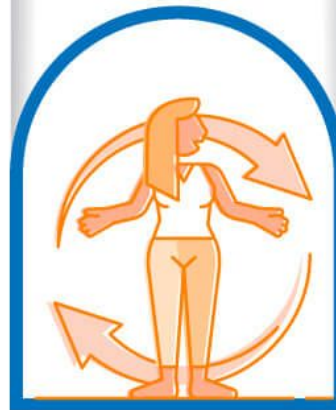
## Proteins are required for muscle health

support muscle contraction & movement



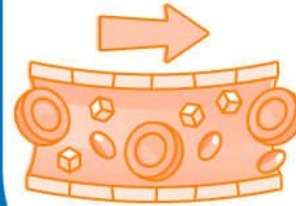
## Proteins are building blocks of human body

provide support to the body



## Proteins are required for making hormones

hormones help coordinate bodily function



## Proteins are required for movement of haemoglobin

move essential molecules around the body

## Proteins support

support the regulation and expression of DNA and RNA



## Proteins are required for making antibodies

antibodies support immune function

# What is digestion?

Digestion is the disintegration of the complex nutrients into simple, soluble and assimilable form.

As proteins are too large to absorb, so dietary proteins are hydrolyzed to amino acids by proteolytic enzymes, which can be easily absorbed

**Mouth**

No  
enzymes

**STOMACH**

Pepsin  
Renin

**Pancreas**

Trypsin  
Chymotrypsin  
Elastase  
Carboxypeptidase A  
Carboxypeptidase B

**Intestine**

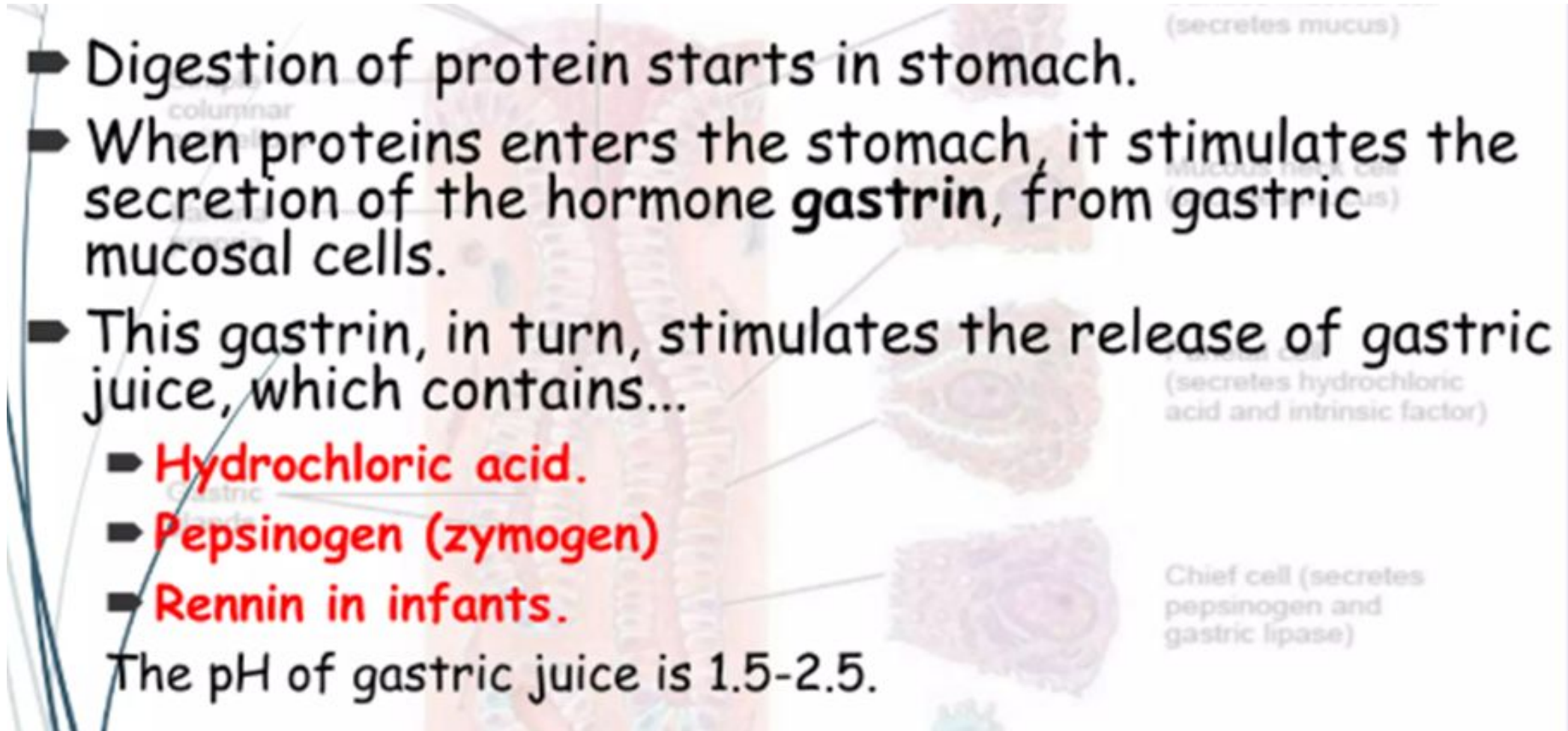
Aminopeptidase  
Dipeptidase

# Digestion In Mouth

- ⌘ No digestion of protein in mouth.
- ⌘ No proteolytic enzymes present in the saliva.
- ⌘ Function of the saliva - lubricate the food, this helps in making food soluble for the action of proteolytic enzymes.
- ⌘ After mastication and chewing, the bolus of food enters stomach where it is acted upon by gastric juice.

# Digestion In Stomach

- Digestion of protein starts in stomach.
  - When proteins enter the stomach, it stimulates the secretion of the hormone **gastrin**, from gastric mucosal cells.
  - This gastrin, in turn, stimulates the release of gastric juice, which contains...
    - **Hydrochloric acid.**
    - **Pepsinogen (zymogen)**
    - **Rennin in infants.**
- The pH of gastric juice is 1.5-2.5.



# Digestion by Gastric Secretion

Digestion of proteins begins in the stomach, which secretes **gastric juice** pH 1.5-2.5 (solution containing HCL & Proenzyme pepsinogen).

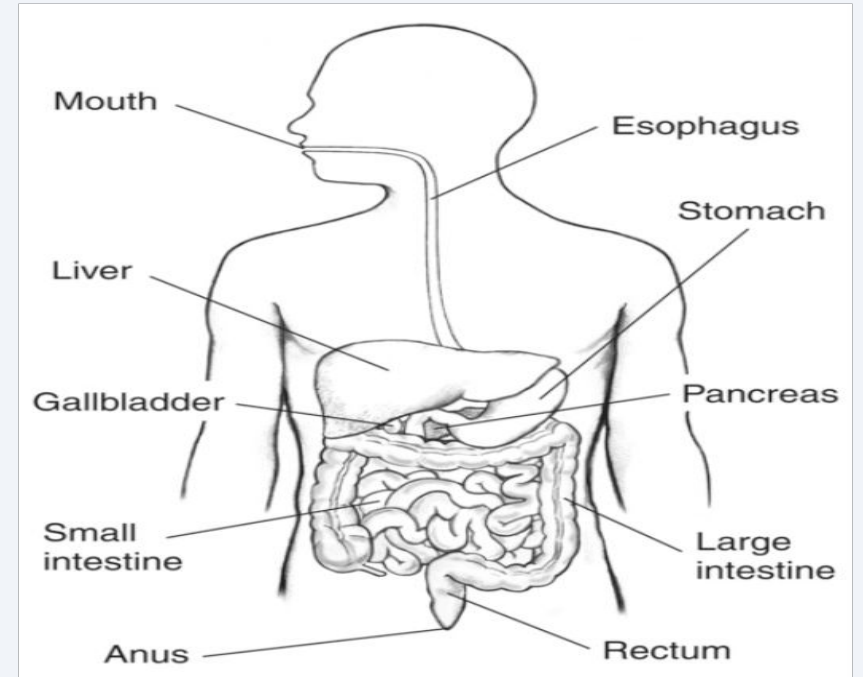
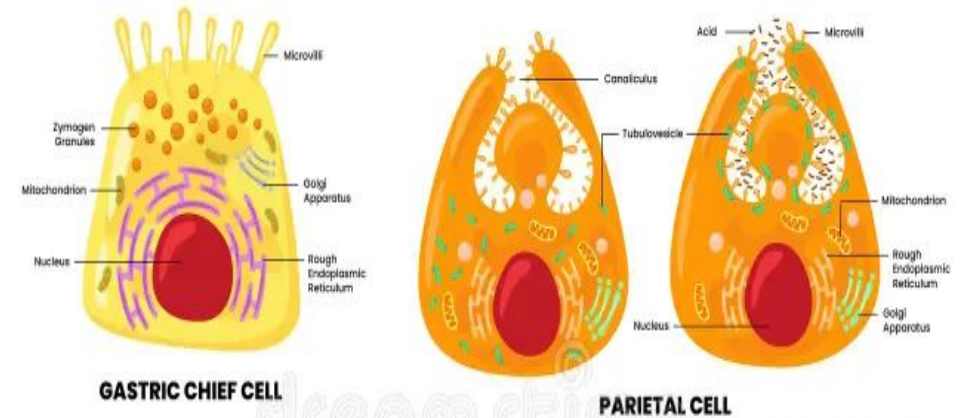
## Hydrochloric Acid (HCl)

- Secreted by **parietal cells** (pH 2.0 – 3.0)
- Kills pathogenic bacteria
- Denatures secondary & tertiary protein structures

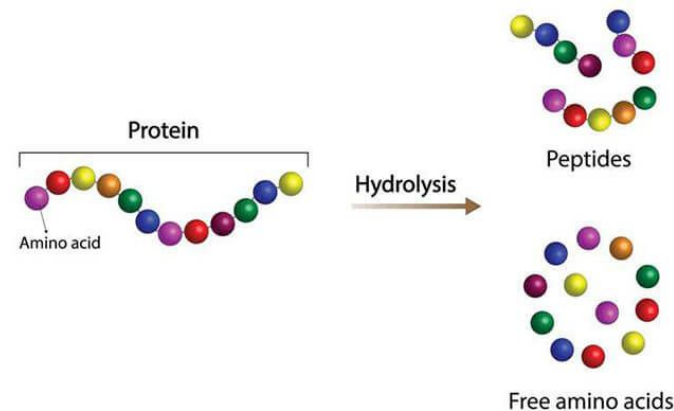
## Pepsin (Acid-Stable Endopeptidase)

- Secreted as inactive zymogen **pepsinogen** by **chief cells**
- In the presence of HCL, pepsinogen undergoes conformational change that allows it to cleave itself (autocatalysis) to active form, pepsin.
- Cleaves internal peptide bonds → yields polypeptides and few free amino acids from dietary proteins

## GASTRIC EPITHELIAL CELLS



| Stomach secrets                                                              | Hydrochloric acid (HCl)                                                                                                                                     | Pepsinogen                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Strong acid<br/><b>(HCl)</b></p> <p>Proenzyme -<br/><b>pepsinogen</b></p> | <p>Denatures proteins</p> <p>Decreases pH (pH 2-3)</p> <p>Activates pepsinogen</p> <p>Kills some bacteria</p> <p>Helps in the absorption of Vitamin B12</p> | <p>Secreted by Chief cells. Activated to pepsin by HCl. And autocatalysis.</p> <p><b>Pepsinogen</b></p> <pre> graph TD     Pepsinogen -- HCL --&gt; Pepsin     Pepsin -.-&gt; Autocatalysis     Autocatalysis --&gt; Pepsin </pre> |



## Pepsin acts on

Peptide bond formed by carboxyl group of Phenylalanine, Tyrosine, Tryptophan, methionine

Converts proteins into proteases and peptones

## RENNIN

Also called chymosin  
Active in infants  
curdling of milk  
Absent in adult stomach

Converts Casein of milk into paracasein  
It combines with calcium to form calcium paracaseinate

# Digestion By Pancreatic Enzymes

- As the acidic stomach contents pass into duodenum, the low pH triggers the secretion of the hormones...
  - **Secretin** - stimulate pancreas to secrete bicarbonate. It neutralizes HCL and rises the pH from 1.5-2.5 to 7.0
  - **Cholecystokinin** - stimulate secretion of pancreatic
    - **Endopeptidase** - Trypsin, Chymotrypsin, Elastase
    - **Exopeptidase** - Carboxypeptidases, Aminopeptidases

## Endopeptidase

- Cleave internal peptide bonds.
- Results into formation of smaller peptides.

Endopeptidase secreted by pancreas are...

- Trypsin (Trypsinogen)
- Chymotrypsin (Chymotrypsinogen),
- Elastase (Proelastase)

## Exopeptidase

- Cleave off 1 AA at time from either the C or N end of the polypeptide

Exopeptidase secreted by pancreas are...

- Carboxypeptidase (Procarboxypeptidase)
- Aminopeptidase

# Pancreatic Neutralization & Proteolytic Cleavage



## Bicarbonate Neutralization ( $\text{HCO}_3^-$ )

Stimulated by secretin; raises intestinal pH to optimize pancreatic enzyme activity.

## Endopeptidases

### Cleave Internal Peptide Bonds

Examples: Trypsin, Chymotrypsin, Elastase

## Exopeptidases

### Cleave Terminal Amino Acids

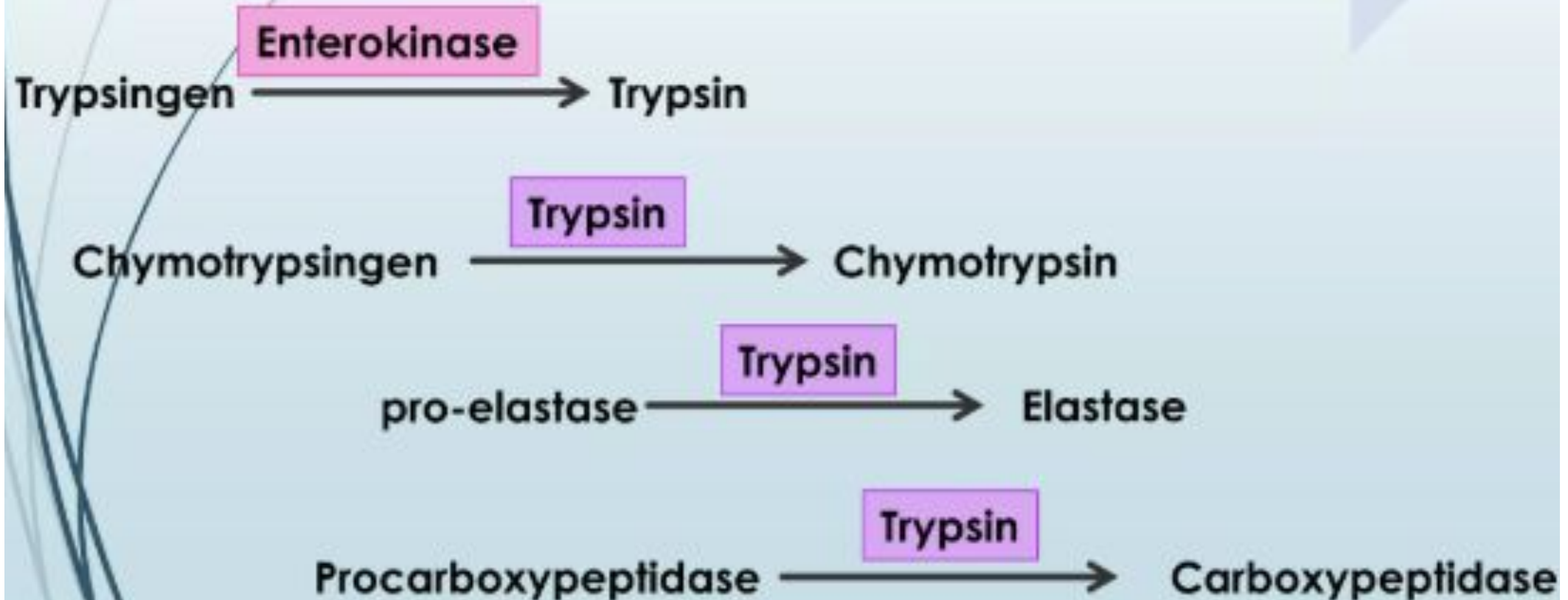
Examples: Carboxypeptidases A & B (C-terminus)

# Pancreatic Protease Cleavage Specificity

| Enzyme             | Type          | Target R-Groups / Cleavage Sites                                                       |
|--------------------|---------------|----------------------------------------------------------------------------------------|
| Trypsin            | Endopeptidase | Arginine (Arg), Lysine (Lys)                                                           |
| Chymotrypsin       | Endopeptidase | Tryptophan (Trp), Tyrosine (Tyr), Phenylalanine (Phe), Methionine (Met), Leucine (Leu) |
| Elastase           | Endopeptidase | Alanine (Ala), Glycine (Gly), Serine (Ser)                                             |
| Carboxypeptidase A | Exopeptidase  | C-terminal Ala, Ile, Leu, Val                                                          |
| Carboxypeptidase B | Exopeptidase  | C-terminal Arg, Lys                                                                    |

# Trypsin

- Secreted in the zymogen form (Trypsinogen)
- **activated by Enterokinase and trypsin itself.**
- specific for cleaving peptide bonds contributed by lysine, arginine (basic amino acids)
- Acts on the zymogen forms of other pancreatic enzymes and activates them
- It has weak action on casein



# Zymogen Activation Cascade

## 1. Enteropeptidase Trigger

Synthesized on mucosal cells; cleaves hexapeptide from Trypsinogen to yield active **Trypsin**.

## 2. Common Activator

Trypsin acts as master activator for all subsequent pancreatic zymogens.

## 3. Activation

Converts Chymotrypsinogen, Proelastase, and Procarboxypeptidase to active enzymes.

## CHYMOTRYPSIN (Endopeptidase)

- ☐ Zymogen form – Chymotrypsinogen
- ☐ activated to chymotrypsin by trypsin
- ☐ Hydrolyzes the peptide bond formed by the carboxyl group of aromatic amino acids

## ELASTASE (Endopeptidase)

- ☐ Zymogen form – Proelastase
- ☐ activated to elastase by trypsin
- ☐ Acts on peptide bonds formed by the amino acids like glycine, alanine, serine

## Carboxypeptidase

- Zinc containing
- Exopeptidase
- Zymogen form is procarboxypeptidase
- Activated by trypsin
- Cannot act on dipeptides

### Carboxypeptidase A

acts on the Carboxy terminal peptide bond connected to Tyrosine, Phenylalanine or tryptophan

### Carboxypeptidase B

acts on the carboxy terminal peptide bond connected to Arginine, lysine

# Digestion By Intestinal Enzymes

- The digestion products of hydrolysis by pepsin, trypsin, elastase, chymotrypsin and carboxypeptidase is completed by the intestinal peptidases, secreted by the mucosa of the small intestine.
- Some of these peptidases are...
  - Aminopeptidase
  - Dipeptidase

## Amino peptidase

- Present on the luminal surface of the intestinal mucosa
- Is an exopeptidase.
- Acts on the amino terminal peptide bond
- Release free amino acid

## Di and Tri peptidases

- Present on the surface of the intestinal mucosal They act on **dipeptides** and release free amino acids
- Enterocytes take up some di and tripeptides
- These peptides are hydrolyzed to amino acids by **intracellular dipeptidase**

# Digestion of Oligopeptides by Surface Enzymes

## Aminopeptidase Location

Exopeptidase bound to the luminal surface of enterocytes (intestinal mucosal cells).

## Cleavage Mechanism

Repeatedly cleaves N-terminal residue from oligopeptides.

### FINAL END PRODUCTS

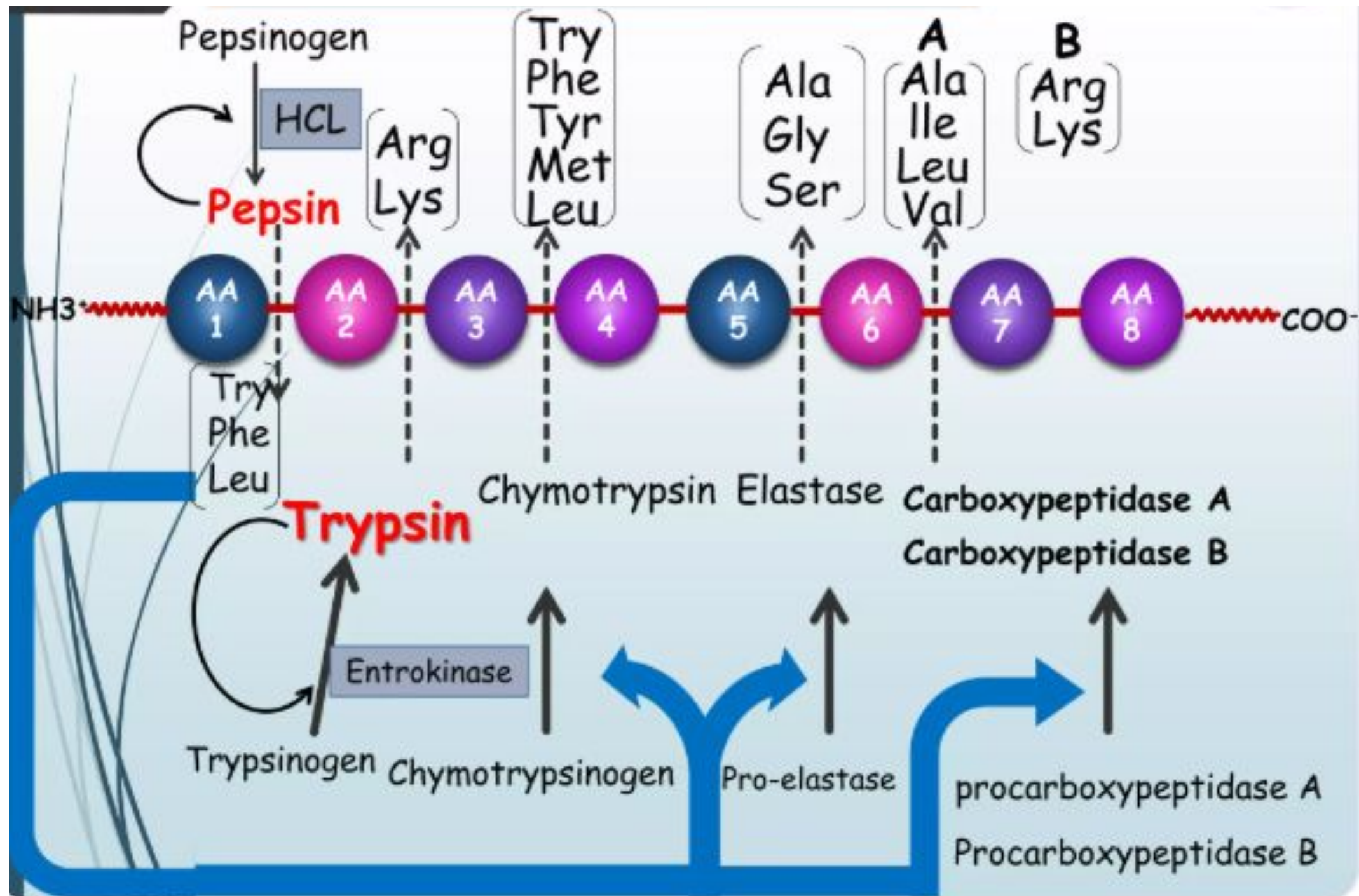
**Free Amino Acids**

+

**Dipeptides**

+

**Tripeptides**



# Intestinal Absorption & Transport Systems



## Free Amino Acids

Na<sup>+</sup>-dependent secondary active transport via Solute Carrier (SLC) proteins.



## Di- & Tripeptides

Taken up via PepT1 (H<sup>+</sup>-linked peptide transporter).

Hydrolyzed internally to free AAs



## Portal Exit & BCAA

Na<sup>+</sup>-independent efflux into portal vein to liver.

BCAAs pass liver directly to muscle

- Absorption of most of the AAs by active transport mechanism.
- By Na<sup>+</sup> dependent active transport system (Na<sup>+</sup> amino acid cotransport)
- An energy requiring process

# Digestive & Secretory Abnormalities

## **Pancreatic Insufficiency**

**Etiology:** Chronic pancreatitis, Cystic Fibrosis, or surgical resection.

**Consequences:** Incomplete digestion/absorption of fats and proteins leading to **steatorrhea** and undigested protein in feces.

## **Celiac Disease (Celiac Sprue)**

**Etiology:** Autoimmune disorder caused by immune response to gluten / gliadin (wheat, barley, rye).

**Consequences:** Immune-mediated damage to small intestine mucosa resulting in severe nutrient malabsorption.

| Transport System                                                                                                                                                                                                | AAs Transported                                                                                                                                                                                                                                                                    | Disorder Associated                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Small neutral amino acids</li> <li>• Large neutral amino acids</li> <li>• Basic amino acids</li> <li>• Acidic amino acids</li> <li>• imino acid and Glycine</li> </ul> | <ul style="list-style-type: none"> <li>• Alanine, Serine &amp; Threonine</li> <li>• Isoleucine, Leucine, Valine, Tyrosine, tryptophane, PA</li> <li>• Arg, Lys, Ornithine &amp; Cystine</li> <li>• Glu acid &amp; Asp acid</li> <li>• Proline, OH-proline &amp; Glycine</li> </ul> | <ul style="list-style-type: none"> <li>• Hartnup disease</li> <li>• Cystinuria</li> <li>• Glycinuria</li> </ul> |

# Cystinuria & COAL Amino Acid Transport Defect

## Incidence & Genetics

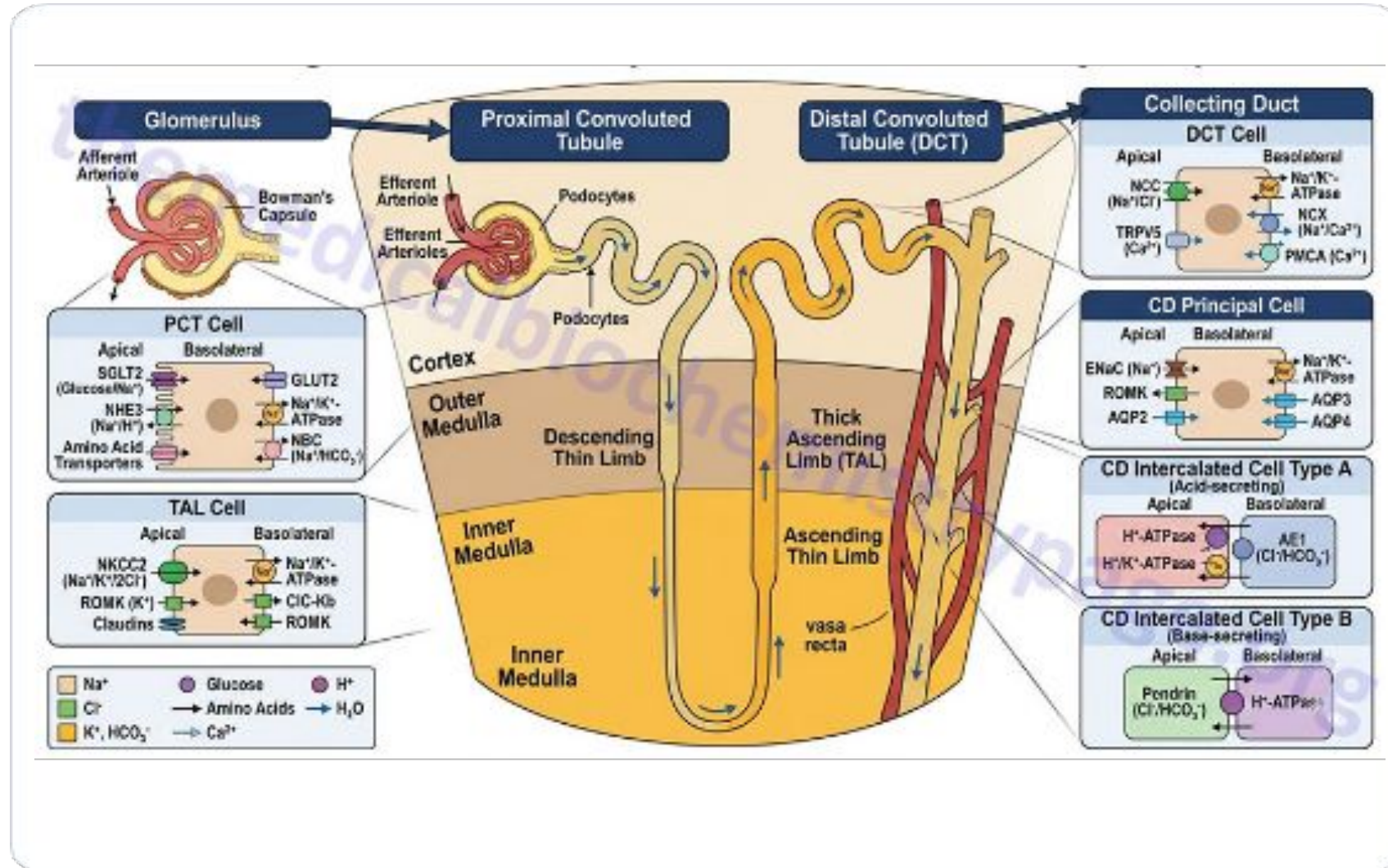
1 in 7,000 individuals — most common inherited disorder of amino acid transport.

## Defective COAL Carrier System

Impairs reabsorption of **Cystine**, **Ornithine**, **Arginine**, and **Lysine** in proximal renal tubules & small intestine.

## Clinical Pathology & Management

Inability to reabsorb cystine leads to accumulation and precipitation of **cystine stones (calculi)** in urinary tract. Managed by oral hydration.



# Hartnup Disorder: Tryptophan Uptake Defect

## Molecular Defect

Inherited defect in the neutral amino acid transporter system, severely reducing intestinal and renal uptake of tryptophan.

## Clinical Manifestation

Tryptophan deficiency leads to reduced Niacin synthesis, resulting in **pellagra-like dermatologic and neurologic symptoms**.

**Comparison: Cystinuria affects COAL amino acids (basic/cystine) → kidney stones; Hartnup affects neutral amino acids (tryptophan) → pellagra-like symptoms.**

THANK YOU

